MPG No. 26.1.95 $\bar{6}m2$ (-6m2 setting) [Type I, hexagonal] [M tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss). * Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzx} & 0 & 0 \\ 0 & -M_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzx} = 3\mathbb{T}_{u}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & 0 & 0 \\ 0 & M_{yzx} & 0 \\ 0 & 0 & M_{xyz} \end{bmatrix}$$

$$M_{yzx} = \mathbb{T}_{0}^{(1)} - \mathbb{T}_{u}^{(2)}$$
$$M_{xyz} = \mathbb{T}_{0}^{(1)} + 2\mathbb{T}_{u}^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & -M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -M_{xxzx} \\ M_{xxzx} & -M_{xxzx} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -M_{xxzx} & 0 & 0 \end{bmatrix}$$

$$M_{xxzx} = \mathbb{M}_{4c}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & -M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{xxzx} \\ -M_{xxzx} & M_{xxzx} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -M_{xxzx} & 0 & 0 \end{bmatrix}$$

$$M_{xxzx} = -2\mathbb{T}_{3s}^{(1)}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & M_{xxzx} & 0 \\ 0 & -M_{xxzx} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{xxzx} & 0 & 0 \end{bmatrix}$$

$$M_{xxzx} = \mathbb{T}_{3s}^{(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzxy} = \mathbb{T}^{(3)}_{3s}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & 0 & 0 & M_{xxxz} & 0 \\ 0 & 0 & 0 & 0 & -M_{xxzx} & 0 & 0 & -M_{xxxz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzxy} \\ -M_{yzxy} & M_{yzxy} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -M_{xxxz} & 0 & 0 & -M_{xxzx} & 0 & 0 \end{bmatrix}$$

$$M_{xxzx} = \mathbb{M}^{(1)}_{4c} - 2\mathbb{T}^{(1)}_{3s} + \mathbb{T}^{(2)}_{3s}$$
$$M_{xxxz} = \mathbb{M}^{(1)}_{4c} - 2\mathbb{T}^{(1)}_{3s} - \mathbb{T}^{(2)}_{3s}$$
$$M_{yzxy} = -\mathbb{M}^{(1)}_{4c} - 2\mathbb{T}^{(1)}_{3s}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzxy} = \mathbb{T}^{(3)}_{3s}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & M_{xxxz} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ -M_{xxxz} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{xxxz} & 0 & 0 \\ 0 & 0 & -M_{xxxz} \\ 0 & -M_{xxxz} & 0 \end{bmatrix}$$

$$M_{xxxz} = \mathbb{M}^{(1)}_{4c}$$